CONCEPT LEARNING FROM JAPANESE COPULAR SENTENCES USING HEURISTICS

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ABSTRACT

This paper presents a method for concept learning from Japanese copular sentences. Copular sentences frequently appear in Japanese text. Our proposed method is based on heuristics by which human beings get semantic concepts from copular sentences. Our proposed method consists of the analysis of sentence structure, learning process, path reduction process, inheritance process and evaluation learning process. The system developed by this method represents the concepts of noun phrases in copular sentences and semantic relations between them in the concept relation network. The system represents unknown concepts with ambiguity in it. However the ambiguity reduces as the system learns from other copular sentences. We developed an experimental system, and carried out an evaluation experiment. In this paper, we outline our method and describe the results of the evaluation experiment.

1. Introduction

It is necessary to develop a natural language understanding system in order to use computers and information more effectively. However, there are still a lot of problems with general and practical use of it. Therefore we choose to research the semantic processing of the copular sentence as a basis for developing it. The copular sentence appears in Japanese text frequently, and consists of a subject and a predicate. The subject and the predicate consist of nouns or noun phrases[1]. We hope that our research could be a basic technic to develop the natural language understanding. system Some researchers have already presented some kinds of research on copular sentences from the point of view of Linguistics[1]. However there has been little research from the point of view of Computational Linguistics. We have done research on the method of learning and analysis for concepts of noun phrases in copular sentences and the semantic relations between them[2,3]. The method of the lexicon acquisition has been presented[4]. This method learns from corpus by capitalizing on lexical categories. However our proposed method uses only raw sentences and heuristics for learning. Therefore the knowledge learned by our method is less than that learned by the method using lexical categories. However, in the real world, we can get only raw sentences. We consider that our method is effective for resolving the mechanism that a human being learns the concepts from the raw sentences in the real world.

This paper describes our proposed method on concept learning from Japanese

copular sentences and the results of evaluation experiment for this method.

2. Semantic Relations of Noun Phrases

The form of the copular sentence is "$< NP1^{*} >$ †*wa* $< NP2 >$ *desu.*"‡ It is translated into "$< NP1 >$ is $< NP2 >$." Basic semantic relations between $< NP1 >$ and $< NP2 >$ are the relations of (1) subclass concept and superclass concept, (2) member concept and class concept, (3) equivalence concept, (4) object concept and property concept, (5) object concept and event concept, (6) object concept and metaphor concept, (7) eel sentence, and (8) equivalence concept with the same words.

The semantic relations of (1), (2), (3), (4) and (5) are literal. The other semantic relations are not literal. In this paper, our proposed method treats only the relations of (1), (2), (3) and (4). The considerations for other relations are future problems of this method.

3. Knowledge Representation and Model of Concept Generation

This method uses the SNePS (Semantic Network Processing System)[5] network as knowledge representation. The SNePS network can express a concept and a relation of concepts in the same format. In the SNePS network, there is a one-to-one correspondence between nodes and represented concepts. Nodes represent intensional objects. Arcs represent structured and semantic relations. Lex arcs connect our intensional world to our external world.

In this paper, the kinds of semantic relations between concepts are limited as Table 1.

Table 1. Symbols of Semantic Relations

Semantic Relations	Symbols	
	SNePS	Heuristics Tables
Subclass and superclass	$\leftarrow sub- : -su \rightarrow$	*sub!su*
Superclass and subclass	$\leftarrow su- : -sub \rightarrow$	*su!sub*
Member and class	$\leftarrow mem- : -class \rightarrow$	*mem!class*
Class and member	$\leftarrow class- : -mem \rightarrow$	*class!mem*
Equivalence	$\leftarrow eq- : -eq \rightarrow$	*eq!eq*
Object and property	$\leftarrow ob- : -prop \rightarrow$	*ob!prop*
Property and object	$\leftarrow prop- : -ob \rightarrow$	*prop!ob*

Figure 1 shows an example of the SNePS network. The "human being" and "mammal" are connected to [*m*1] §and [*m*3] by lex arcs respectively. The "human

*NP represents the noun phrase.

†$< X >$ represents a lexicon of the X.

‡An italic character represents Japanese.

§$[X]$ represents a concept of the X.

being" and "mammal" express lexicons. [$m1$] is the concept of "human being". [$m3$] is the concept of "mammal". [$m2$] is the concept of the semantic relation between [$m1$] and [$m3$] and the proposition of "*Ningen wa honyurui desu.*(A human being is a mammal.)". [m2] and the two arcs represent the subclass and superclass relation between [$m1$] and [$m3$].

A path is a sequence of arcs from a concept to another concept. The path has two kinds. One is a direct path that has only one relation concept. The other is a lengthy path that has more than two relation concepts. In our proposed method, a concept has two guidelines. One guideline is a known and unknown concept. The other guideline is a primitive and complex concept. The known concept is the concept that the SNePS network has already. The unknown concept is the concept that the SNePS network does not have. The lexicon of the primitive concept consists of a noun. The lexicon of the complex concept consists of nouns. Therefore our proposed method has four concept spaces.

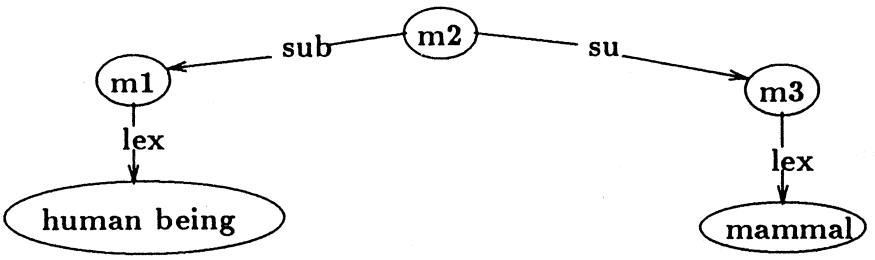


Fig. 1. Example of the SNePS Network

A new concept is generated as an unknown and primitive concept, or as an unknown and complex concept when an unknown concept is inputed. It changes to the known concept when the processing of a sentence is done. In this paper, we call a SNePS network like this, the concept relation network.

4. Method for Concept Learning from Copular Sentences

Figure 2 shows the outline of this method. Our proposed method decides the semantic relation between two concepts on the basis of the paths between them.

4.1. Analysis of Sentence Structure

Noun phrases 1 and 2 are extracted from a copular sentence. In this method, the copular sentence is already divided into morphemes by a space. Therefore the noun phrase 1 is the morpheme between the top and "*wa, ga*". The noun phrase 2 is morpheme between "*wa, ga*" and "*desu, dearu ,da*(inflectional endings)".

In this paper, the noun phrase is limited in the forms of " $< N1$[¶]$>$ *no* $< N2 >$" and " $< N1 >$ *to*(and) $< N2 >$". The *"no"* is a Japanese particle. In " $< N1 >$ *no* $< N2 >$", the relations between the concept of $< N1 >$ and the concept of $< N2 >$ are possession, whole and part, place limitation, time limitation, material limitation,

[¶] N represens the noun.

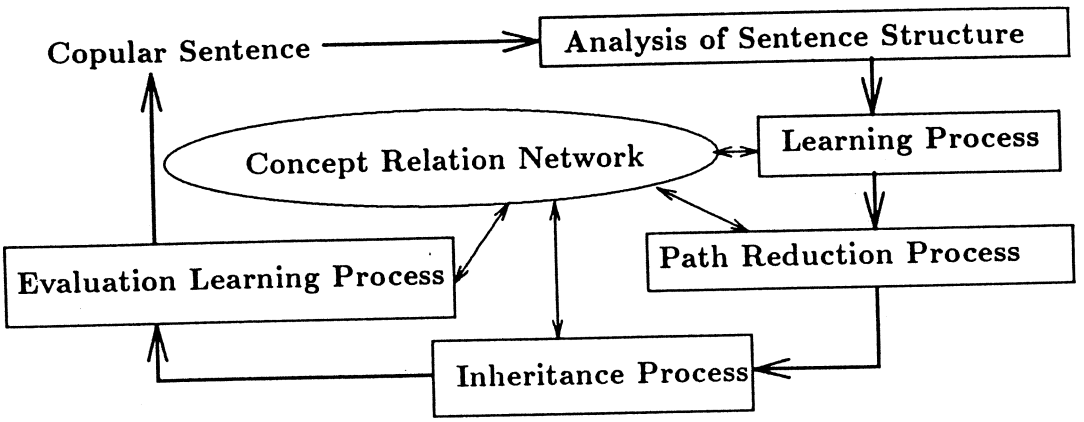


Fig. 2. Outline of Process

kind limitation, quantity limitation, case relations and so on. In this paper, this method treats the possession, the whole and the part, the place limitation, the kind limitation and the case relations. The sentences of meta-knowledge are included in the input copular sentences. They are defined as sentences in which the noun phrase 2 is the lexicon for the definition of the concept relation.

4.2. Learning Process

This process generates unknown concepts and decides semantic relations between concepts by some heuristics. These heuristics have been created comprehensively. The learning heuristics have two kinds. One is based on the structure of the input sentence. The other is based on the situation of the connected concepts. The former has seven kinds. Table 2 shows them. Those are expressed by LHSS (Learning Heuristics based on the Structure of Sentence). The latter has three kinds. Tables 3, 4 and 5 show them. Those are expressed by LHSC (Learning Heuristics based on the Situation of the connected Concepts). In table 2, the first heuristics means that the relations between the concept of $< NP1 >$ and the concept of $< NP2 >$ are limited to the relations of "sub!su", "mem!class", "eq" and "ob!prop" in "$< NP1 > wa < NP2 > desu.(< NP1 > \text{is} < NP2 >.)$"

LHSC is based on the situation of the connected concepts. Figure 3 shows their situation. In this figure, A, B, C1 and C2 are nodes that represent concepts. R, R1 and R2 are arcs that represent semantic relations between concepts. Table 3 shows the heuristics for R based on R1. Table 4 shows the heuristics for R based on R2. Table 5 shows the heuristics for R based on R1 and R2. In table 3, the second column means that R is limited to the relations of "eq" and "prop!ob" in the case that R1 is the relation of "mem!class". In table 4, the first column means that R is limited to the relations of "sub!su", "eq" and "ob!prop" in the case that R2 is the relation of "sub!su". In table 5, the first line and the first column means that R is limited to the relations of "sub!su", "eq" and "ob!mem" that R1 and R2 is the relation of "sub!su" respectively.

Table 2. Learning Heuristics based on the Structure of the Sentence

Structures	Lexicons of Objects	Relations
$< NP1 > wa < NP2 > desu.$ $(< NP1 >$is$< NP2 >.)$	$< NP1 >:< NP2 >$	sub!su mem!class eq ob!prop
$< NP >$is"$< N1 > no < N2 >$."	$< N1 >:< N2 >$ $< N1 > no < N2 >:< N1 >$ $< N1 > no < N2 >:< N2 >$	ob!prop ob!prop sub!su mem!class
$< NP >$is"$< N1 > to < N2 >$."	$< N1 >:< N1 > no < N2 >$ $< N2 >:< N1 > no < N2 >$	sub!su mem!class sub!su mem!class
$< N1 > no < N2 > wa < N3 > desu.$ $(< N1 > no < N2 > is < N3 >.)$	$< N1 > no < N2 >:< N3 >$ $< N2 >:< N3 >$	eq su!sub class!mem
$< N1 > to < N2 > wa < N3 > desu.$ $(< N1 >$and$< N2 >$is$< N3 >.)$	$< N1 > to < N2 >:< N3 >$ $< N1 >:< N3 >$ $< N2 >:< N3 >$	sub!su eq sub!su mem!class sub!su mem!class
$< N1 > wa < N2 > no < N3 > desu.$ $(< N1 >$is $< N2 >no< N3 >.)$	$< N1 >:< N2 > no < N3 >$ $< N1 >:< N3 >$	sub!su mem!class eq sub!su mem!class
$< N1 > wa < N2 > to < N3 > desu.$ $(< N1 >$is$< N2 >$and$< N3 >.)$	$< N1 >:< N2 > to < N3 >$ $< N1 >:< N2 >$ $< N1 >:< N3 >$	su!sub class!mem su!sub class!mem su!sub class!mem

Table 3. LHSC : R based on R1

R1	sub!su	mem!class	su!sub	class!mem	eq	ob!prop	prop!ob
R	–	eq prop!ob	–	–	–	sub!su mem!class eq	–

"-" represents that R is not limited.

Table 4. LHSC : R based on R2

R2	sub!su	mem!class	su!sub	class!mem	eq	ob!prop	prop!ob
R	sub!su eq ob!prop	mem!class eq ob!prop	sub!su eq ob!prop	sub!su eq ob!prop	–	–	sub!su eq

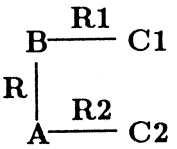


Fig. 3. Situation of the Connected Concepts

Table 5. LHSC : R based on R1 and R2

R2	R1						
	sub!su	mem!class	su!sub	class!mem	eq	ob!prop	prop!ob
sub!su	sub!su eq ob!mem	X1	sub!su eq ob!prop	sub!su eq ob!prop	sub!su eq ob!prop	sub!su eq	sub!su eq ob!prop
mem!class	mem!class ob!prop	eq	mem!class ob!prop	mem!class ob!prop	mem!class eq ob!prop	mem!class eq	ob!prop
su!sub	sub!su eq ob!prop	X1	sub!su eq ob!prop	sub!su eq ob!prop	sub!su eq ob!prop	sub!su eq	sub!su eq ob!prop
class!mem	sub!su eq ob!prop	X1	sub!su eq ob!prop	sub!su eq ob!prop	sub!su eq ob!prop	sub!su eq	sub!su eq ob!prop
eq	–	X2	–	–	–	sub!su mem!class eq	–
ob!prop	–	X2	–	–	–	sub!su mem!class eq	–
prop!ob	sub!su eq	X1	sub!su eq	sub!su eq	sub!su eq	X3	sub!su eq

X1 represents no relation or "class!mem". X2 represents no relation or "eq".
X3 represents no relation or "prop!ob".

4.3. Path Reduction Process

Path reduction is defined as converting a lengthy path into the equivalently direct path. Path reduction rules are used in the path reduction process[3]. They are automatically generated on the basis of the situation of the connected concepts. This process gives not only the reduction of ambiguity but also generality to this method.

The left-hand side of the path reduction rule is the specific lengthy path and the right-hand side is the direct path without ambiguity. An example of the path reduction rule is as follows:

$$[mX] \leftarrow sub!su \rightarrow [mY] \leftarrow sub!su, mem!class \rightarrow [mZ]$$
$$\Downarrow$$
$$[mX] \leftarrow sub!su \rightarrow [mZ]$$

4.4. Inheritance Process

The property of the concept of " $< N1 >$ *no* $< N2 >$ " depends on that of $[N1]$. Therefore, we consider that the property inheritance from the concept of " $< N1 >$ *no* $< N2 >$ " to $[N1]$ is possible. The inheritance process of this method treats only " $< N1 >$ *no* $< N2 >$ ". A new concept is generated when the situation of the concept relation network satisfies the inheritance heuristics[3].

4.5. Evaluation Learning Process

This process recovers an erroneous semantic relation that has already been decided without ambiguity. The errors occur by reason of a change of situation and erroneous applications of the heuristics. This process has some heuristics[2].

5. Evaluation Experiment

5.1. Data and Procedure

Fifty copular sentences are inputed to the experimental system developed on the basis of our method. The initial world for this experiment includes the network of figure 1, the concepts and the lexicons of "male" and "sex", and the concept of the semantic relation "mem!class" between these two concepts. All these concepts are known and primitive concepts.

5.2. Results and Considerations

Path reduction rules were generated in seven sentences and applied in six sentences. The inheritance process was performed in two sentences. Figure 4 shows the change of ambiguity of the concept relation network. Ambiguity degree is expressed by AMD (AMbiguity Degree). AMD is defined as follows:

$$AMD = \frac{\text{Total number of relations}}{\text{Total number of arcs}} \tag{1}$$

The AMD was reduced to about 1.3 after learning from fifteen sentences. This result shows that this method is effective for the reduction of ambiguity.

In the world after learning from fifty sentences, all relations decided without ambiguity are correct and all relations with ambiguity include a correct relation respectively.

6. Conclusions

In this paper, we proposed the method for learning concepts found in copular sentences and relations between them. This method gives a basis for the realization of natural language understanding system. Our proposed method is especially effective for the reduction of ambiguity. This method uses some heuristics and the path reduction rules. We developed the experimental system on the basis of this method and have done the evaluation experiment. Our result was that the ambiguity degree was

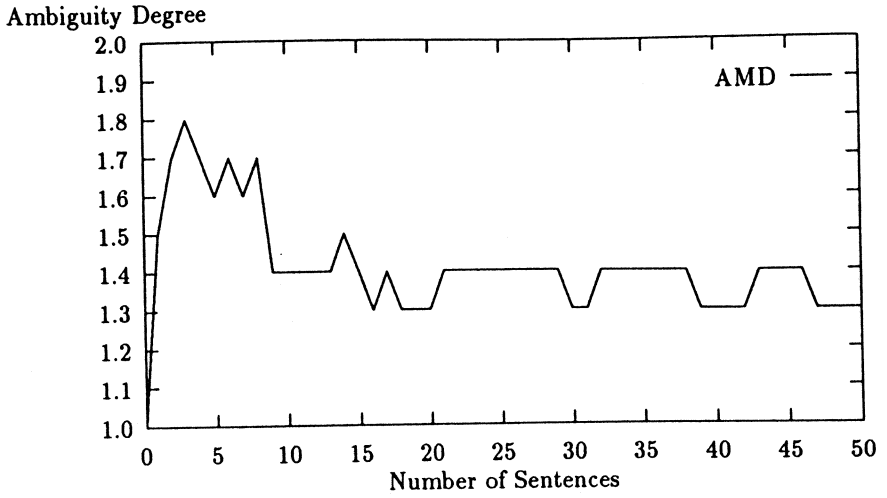


Fig. 4. Change of Ambiguity Degree

reduced to about 1.3 after learning from fifty sentences. All relations are correct or include a correct relation respectively in the world after learning from fifty sentences. This results show that this method is effective as the basis of developing a natural language understanding system.

Future problems for this method include the extension for other structures of a noun phrase and the extension of other parts of speech. The goal of this research is that this method can learn from standard sentences by using this approach.

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